



PRESENTER SCRIPT · 90 MINUTES

INTRO TO SYNTHESIS

One beat per slide, with a running clock. Every beat names the slide you put up, then what you do over it: SLIDE / SAY / DEMO / THEY DO. 21 September 2026 · Club Control, Bucharest · Sintezaur × Zeedo · powered by Sequential.

BEFORE DOORS

- All synths powered on, **main** at 12 o'clock, sound in both headphones at every station, Basic Program loaded on all of them (hold **transpose down** + **program**).
 - **On the loaner units, go through the GLOBAL menu.** Each of these breaks an exercise if it is set the other way:
 - **Pot Mode = Jump** (Global 4). The one that matters most. Every exercise says "cutoff to 9 o'clock"; on any other setting the knob does nothing until it reaches the value stored in the program, and the person turning it thinks they broke the synth.
 - **Local Control = On** (Global 3). On Off the keyboard stops talking to the synth: keys press, nothing sounds.
 - **Aftertouch = Poly** (Global 9) and **Atouch Curve = Medium** (Global 10). Module 07 is about the pressure on each separate key; on mono there is nothing to show. Medium is the straight curve.
 - **Velocity Curve = Medium**, for the same reason.
 - **Clock Mode = Out or Off**. On In or In/Thru the arpeggiator waits for a clock from outside and simply never starts, which stops module 08 dead.
 - **Alt Scale = Equal Temperament**, **Coarse Tune** and **Fine Tune = 0**, so every instrument in the room is tuned the same.
 - **Tune Voices**, after fifteen minutes of warm-up.
- Keybed firmware: **keymech OS 1.0.0.2 or newer**. On older versions the lowest seven keys send maximum aftertouch by themselves. After any update run the aftertouch calibration, then press into those low keys and check that they respond gradually.
- Presenter's synth → PA (main out, mono 1/4"), microphone tested from the back of the room; 16:9 screen, the RO deck open on the title slide, clicker; overhead camera framed on the panel (if available).
 - On your own synth: the "wow" preset for 00-03 and "Glass Echo" (F1 17) for 06-05, both tested; the AT → LFO amount test done at home (its result decides the wording in 07-04).
 - student handbooks, if they are printed: one on every station (RO; EN on request), the rest by the exit; water; photo/video.
 - Presets to audition beforehand, for the DEMOs (names from the F1 list; check the sound yourself): 05 Syncing Pressure, 07 Mod My Arp, 17 Glass Echo, 18 Uptight Bass, 25 Dynamik Pad, 30 Unisyncerizer, 46 Acids Pro, 77 Poseidon PWM, 93 Hard Sync, 112 Aggro Feedback, 118 Bottle Service.

PLAN B

- **No projector:** teach from the panel; if the handbook is printed, its pages follow the slide order; overhead camera on a TV, if one exists.
- **No PA:** participants put their headphones down and you play your synth through an amp or a local speaker; demos get cut to half length.
- **A synth freezes or sounds wrong:** reset to Basic Program (hold **transpose down** + **program**); if it persists, check **main**, the mixer (at least one source up), **cutoff**, the amounts at 12 o'clock, **key track**; last resort: swap that station with yours.

0:00–0:05

00·WELCOME & SETUP

00·01 TITLE · 00·02 TODAY · 00·03 DEMO · THE DISTANCE · 00·04 FIRST CONTACT

0:05–0:12

01·SIGNAL FLOW

01·01 COVER · 01·02 WHAT'S IN A SOUND · 01·03 THE VOICE PATH · 01·04 WHERE ON FOURM · 01·05 THE MIXER · 01·06 TRY IT · 01·07 KEY TERMS

0:12–0:23

02·OSCILLATORS

02·01 COVER · 02·02 THE WAVESHAPES · 02·03 OCTAVES & TUNING · 02·04 PULSE WIDTH & PWM · 02·05 THE SUB TRICK · 02·06 NOISE · 02·07 TRY IT · 02·08 KEY TERMS

0:23–0:33

03·FILTER

03·01 COVER · 03·02 LOW-PASS · 03·03 RESONANCE · 03·04 KEY TRACK & ENV AMOUNT · 03·05 TRY IT · 03·06 KEY TERMS

0:33–0:44

04·ENVELOPES

04·01 COVER · 04·02 ADSR · 04·03 TWO ENVELOPES ON FOURM · 04·04 BASS vs PAD · 04·05 TRY IT · 04·06 KEY TERMS

0:44–0:53

05·LFO & MODULATION

05·01 COVER · 05·02 THE LFO · 05·03 THE MODULATION SECTION · 05·04 TRY IT · 05·05 KEY TERMS

0:53–1:02

06·VOICES & UNISON

06·01 COVER · 06·02 POLYPHONY · 06·03 UNISON · 06·04 CHORD MEMORY · 06·05 FEEDBACK · 06·06 TRY IT · 06·07 KEY TERMS

1:02–1:10

07·EXPRESSION

07·01 COVER · 07·02 VELOCITY · 07·03 POLY AFTERTOUCH · 07·04 PRESSURE ROUTING · 07·05 GLIDE · 07·06 TRY IT · 07·07 KEY TERMS

1:10–1:17

08·ARPEGGIATOR & SEQUENCER

08·01 COVER · 08·02 HOLD A CHORD · 08·03 CLOCK & SYNC · 08·04 NOTE & MOD SEQ · 08·05 TRY IT · 08·06 KEY TERMS

1:17–1:30

09·WRAP-UP · Q&A · FREE PLAY

09·01 THE ROAD, AGAIN · 09·02 CHALLENGE · 09·03 Q&A · 09·04 THANK YOU · 09·05 FREE PLAY

0:00–0:05

MODULE 00·WELCOME & SETUP

0:00–0:01

00·01 TITLE Headphones down

SLIDE 1 · COURSE MAP (HUB)

SAY: “Welcome. I'm Iulian, from Sinteaur. Over the next ninety minutes you'll come to understand every stage a sound passes through inside a subtractive synthesizer, and you'll hear it, not from me, but from under your own fingers, on the instrument in front of you: the Sequential Fourm. Thank you to Zeedo, who made this possible, and to Sequential, who lent us the instruments. The map behind me is the whole course: eight stops, in the order the signal travels.”

0:01–0:03

00·02 TODAY

SLIDE 2 · TODAY

SAY: “Three things before we start. One: you're two to a synth. Swap at the controls at every exercise, so the panel passes through both pairs of hands, and both of you watch me: nobody has to read anything. Two: headphones, on when you play, one ear free when I'm talking; MAIN volume at twelve o'clock, and we leave it there, the one exception is when I ask you to lower it, before the filter starts singing on its own. Three: if you get lost, you have a home button: hold TRANSPPOSE DOWN

and press PROGRAM. That loads Basic Program, a plain saw with the filter open. Pressing PROGRAM on its own is harmless; what loses your sound is turning SELECT while the display shows the program list, that loads another program. And: there are no wrong sounds today. Explore first, then we talk.”

0:03-0:04 00-03 DEMO · THE DISTANCE PA · Headphones down

SAY: “This is a factory sound.” (play the spectacular preset for 20 seconds) “And this is point zero.” (Basic Program, a few notes) “By the end we’ll have built the distance between these two, and you’ll know what every knob does along the way.”

0:04-0:05 00-04 FIRST CONTACT

SLIDE 3 · WHERE · BASIC PROGRAM

SAY: “Headphones on. Play a few notes on whatever is loaded right now. Then do the reset: hold TRANSPOSE DOWN, press PROGRAM. Play again: this plain saw is where every sound today starts from. Swap and repeat.”

THEY DO: (A) plays a few notes → hold **transpose down** + press **program** → plays again. (B) the same. You walk the tables and check everyone is on Basic Program.

0:05-0:12 MODULE 01 · SIGNAL FLOW

0:05-0:06 01-01 COVER Headphones down

SLIDE 4 · COVER

SAY: “The whole course rests on a single idea: sound follows a path. Learn the path once and no synthesizer will ever look complicated again, no matter how many knobs it has.”

0:06-0:07 01-02 WHAT'S IN A SOUND PA · Headphones down

SLIDE 5 · WHAT'S IN A SOUND

SAY: “What’s inside a sound? A note isn’t a single frequency: it’s one base frequency, the fundamental, plus a tower of quieter tones stacked above it: the harmonics. Their blend is what we hear as timbre. A saw has all of them, a square only the odd ones, a triangle the odd ones too but fading fast, which is why one is bright, one hollow, one soft. Subtractive synthesis starts from a rich wave and takes away. Think of a sculptor: the raw block is the wave, the chisel is the filter, and the shape over time comes from the envelopes.”

DEMO: (20 s): a lone saw, “a boring buzz, like the cheapest transistor organ”, then the same saw with the filter half closed and a slow attack: “the same block, sculpted.”

0:07-0:08 01-03 THE VOICE PATH

SLIDE 6 · VOICE PATH

SAY: “Left to right: the oscillators make the raw tone, the mixer balances four sources, the filter sculpts the brightness, the amplifier shapes the volume, the speaker gets the result. The dotted line: the Fourm can feed its own signal back into the mixer: feedback; we’ll use it in module 6.”

0:08-0:09 01-04 WHERE ON FOURM

SLIDE 7 · WHERE ON FOURM

SAY: “Here’s the best part: the panel is laid out in the same order as the path. Look at your synth and put a finger on it: oscillators, mixer, filter, envelopes, main. The sections on the left, modulation, aftertouch, and the ones along the bottom, unison, glide, LFO, clock, arpeggiator, we’ll take one by one, in the modules ahead.”

0:09-0:10 01-05 THE MIXER PA

SLIDE 8 · THE MIXER

SAY: “Four sources: Osc A, Osc B, noise, feedback. The golden rule of troubleshooting, which I’ll repeat all evening: if you hear nothing, check the mixer first, then the cutoff, then any amount knob left at twelve o’clock.”

DEMO: bring up Osc A alone, add B, then a hiss of noise on top.

0:10-0:12 01-06 TRY IT

SAY: “Headphones on. From Basic Program: hold a note and take OSC A in the mixer down to zero. Silence: no source, no sound. Bring it back. Make sure SAW on Oscillator B is lit (press it if it’s dark) and raise OSC B: two oscillators, one note, already fatter. Swap: your partner does the same, and at the end, trace the path across the panel with a finger.”

THEY DO: (A) **osc a** to zero → silence → back up; Osc B **saw** lit; **osc b** up. (B) repeats; both trace the path with a finger. You walk the tables.

0:12 01-07 KEY TERMS

SLIDE 9 · KEY TERMS

SAY: “The terms: VCO, mixer, VCF, VCA, feedback, harmonics. They come back in every module; stop me now if any of them is unclear.” A real three-second pause.

0:12-0:23

MODULE 02 · OSCILLATORS

0:12-0:13 02-01 COVER Headphones down

SLIDE 10 · COVER

SAY: “First stop: the place where sound is born. Everything further down the path only removes or reshapes what the oscillators give us. That’s why the waveshapes matter so much.”

0:13-0:15 02-02 THE WAVESHAPES PA · Headphones down

SLIDE 11 · WAVESHAPES

SAY: “Three shapes, three personalities. Saw: all the harmonics, the bright workhorse, brass, strings. Triangle: soft, almost pure, the sub layer. Pulse: hollow like a clarinet when it’s square, thin and nasal as it narrows, with a secret parameter we’ll open up in a moment.”

DEMO: same note, slowly switch through the three shapes. Then the game (30 s): “Which shape is this?”: play saw / pulse / tri at random, the room shouts. “Learn to recognize them and you’ll hear them in every track from now on.”

0:15-0:16 02-03 OCTAVES & TUNING

SLIDE 12 · OCTAVES & TUNING

SAY: “The registers: five octave positions, from minus two to plus two, on each oscillator; A and B an octave apart means instant size. FREQUENCY is the fine tuning, up to seven semitones. And here’s the first secret of ‘fatness’: detune B slightly against A and the two waves start beating against each other: the sound breathes. Too far, and you hear two notes.”

SLIDE 13 · WHERE · OSC B › OCTAVE · FREQUENCY

DEMO: B’s **frequency** from center slowly to the left: slow beats → chorus → two notes → back.

0:16-0:17 02-04 PULSE WIDTH & PWM

SLIDE 14 · PULSE WIDTH

SAY: “The pulse’s secret: its width. On the Fourm there’s no knob: hold PULSE and turn SELECT. Zero is the square; toward the edge it thins out, like through a straw, until it disappears. Hand its width to the LFO and the tone won’t sit still: a single oscillator that sounds like a chorus. That’s PWM, and we’ll do it in module 5.”

SLIDE 15 · WHERE · OSC A › PULSE|PW + SELECT

DEMO: sweep the PW by hand.

0:17-0:18 02-05 THE SUB TRICK PA

SLIDE 16 • SUB TRICK

SAY: “The ten-second trick: Osc B on triangle, one octave below A. The nearly pure wave slides underneath the tone and adds nothing but weight, no clutter.”

DEMO: A alone, add the sub, take it out: “the room has to feel the hole.”

0:18-0:19 02-06 NOISE PA

SLIDE 17 • NOISE

SAY: “Noise: every frequency at once, no pitch, the raw material of drums. Three colors: white is full, pink is warm, violet leans into the highs. Press NOISE, hold and turn SELECT for the level. A little sharpens the attack; a lot becomes percussion.”

DEMO: (20 s): noise alone with a short envelope = hi-hat.

0:19-0:22 02-07 TRY IT You walk the tables

SAY: “Headphones on, from Basic Program. Osc A: press PULSE, so both saw and pulse are lit. Hold PULSE and turn SELECT toward the edge, then back. Osc B: make sure SAW is lit, octave at zero, raise OSC B in the mixer. Now nudge B's FREQUENCY a little off centre and listen to the beats; too far and it splits into two notes, bring it back to the click in the middle. Swap. Partner: press SAW on B to switch it off, press TRI, octave to minus one, put the sub in, take it out, feel the weight. Press NOISE (it comes in loud) hold it and turn SELECT down to about thirty: just a bit of air.”

THEY DO: (A) Osc A: **pulse** on (saw + pulse) → hold **pulse|pw** + **select** toward the edge and back → Osc B: **saw** lit, octave 0, **osc b** up → B's **frequency** slightly off-centre → back to the detent. Swap. (B) Osc B: **saw** off, **tri** on, octave -1 → **osc b** down / up → **noise** on → hold **noise** + **select** down to ~30 → plays. The result stays on the synth: it's the raw material for the bass in module 4.

0:22-0:23 02-08 KEY TERMS

SLIDE 18 • TERMS

SAY: “Recap: waveshape, harmonics, pulse width and PWM, sub, noise, the LO modes: for the curious, they're in the student handbook. Question for the room: who has heard PWM in a track without knowing its name? Everyone, starting today.”

0:23-0:33 MODULE 03 • FILTER

0:23-0:24 03-01 COVER Headphones down

SLIDE 19 • COVER

SAY: “The oscillators gave us the block of harmonics. The filter is the chisel. It's the most important knob on any synthesizer, and the one you'll touch the most tonight.”

0:24-0:26 03-02 LOW-PASS PA · Headphones down

SLIDE 20 • LOW-PASS

SAY: “Low-pass means exactly what it says: what sits below a certain frequency passes, what sits above fades away. That frequency is the cutoff. Close it, darker; open it, brighter. It's like a trumpet player's hand over the bell: cover it and the sound turns softer, rounder, more muffled. Fully closed, nothing gets through: that's why ‘I can't hear anything’ almost always starts at the cutoff.”

SLIDE 21 • WHERE • FILTER › CUTOFF

DEMO: held chord, cutoff from open to closed and back, slowly.

0:26-0:28 03-03 RESONANCE PA

SLIDE 22 • RESONANCE

SAY: “Resonance is an accent right at the cutoff. Picture a string stretched in front of a speaker: it only vibrates on its own when you hear its note: that’s resonance. A little gives focus and a vowel-like character when you move the cutoff: eee-yah-ooo-uuu” (sing the sweep with the room) “A lot makes the filter whistle alongside the note. At maximum, the filter feeds itself and produces a clean sine with no oscillator at all, like a microphone too close to a speaker. That is called self-oscillation. A fifth sound source.”

SLIDE 23 • WHERE • FILTER › RESONANCE

DEMO: 1: sweep with resonance at three quarters. **DEMO 2 (30 s), “the filter becomes an oscillator”:** oscillators at zero in the mixer, resonance max, key track full → play a melody on the filter’s sine. Warning: “it comes in sudden and loud, bring the volume down a touch first.”

0:28–0:29 03-04 KEY TRACK & ENV AMOUNT

SLIDE 24 • KEY TRACK AND ENV AMOUNT

SAY: “Two small things with a big effect. Key track: the cutoff follows the keyboard: higher notes get a more open filter, the way acoustic instruments behave; without it, the top notes vanish behind the filter. Press KEY TRACK to switch it on (the LED lights) then hold it and choose the amount with SELECT: Quarter, Half, Full. With full tracking, the self-oscillating filter plays a tuned scale. And ENV AMT: how much the filter envelope moves the cutoff over the life of the note, plus or minus, with zero at twelve o’clock. That’s the story of the next module.”

SLIDE 25 • WHERE • FILTER › KEY TRACK • ENV AMT

DEMO: Optional (20 s), “a kick from nothing”: same as before + filter env with a short decay → a kick drum out of the filter.

0:29–0:32 03-05 TRY IT You walk the tables

SAY: “Headphones on. Hold a chord. Cutoff slowly down until it’s dark, then back. Resonance to three o’clock and one more sweep: the whistle rides the sweep. Now, careful: bring MAIN down a little, resonance to maximum, close the cutoff a bit and play single notes: the filter self-oscillates: it sings on its own, at one fixed pitch. Press KEY TRACK so the LED lights, hold it and turn SELECT to Full, then play a scale: the whistle follows the keys, in tune. Resonance back to nine o’clock, play low and high, the top notes come out brighter. MAIN back to twelve, cutoff to twelve, we keep this for the next module. Swap in the middle.”

THEY DO: (A) **cutoff** down → up; **resonance** at 3 o’clock, sweep. Swap. (B) **main** slightly down → **resonance** max → **cutoff** a bit closed → single notes → **key track** on (LED) → hold **key track** + **select** → Full → scale → **resonance** 9 o’clock → high/low notes → **main** 12 o’clock, **cutoff** 12 o’clock.

0:32–0:33 03-06 KEY TERMS

SLIDE 26 • KEY TERMS

SAY: “Cutoff, resonance, self-oscillation, key track, env amount. Five words you’ll use on every synth in the world. And a line worth remembering: run a Moog oscillator through a Korg filter and it sounds like a Korg: the filter carries the identity.”

0:33–0:44

MODULE 04 • ENVELOPES

0:33–0:34 04-01 COVER Headphones down

SLIDE 27 • COVER

SAY: “An oscillator through a filter is still a beep: it starts when you press, stops when you let go, and nothing happens in between. Real instruments have a shape in time. An organ starts and stops instantly. A plucked string starts loud and dies away. A bow swells into the note. The envelope is that shape, drawn with four knobs and replayed on every key.”

0:34–0:36 04-02 ADSR PA

SLIDE 28 • ADSR

SAY: “Imagine turning a stereo’s volume knob over the course of a single note: from zero to ten, then down to five, hold it there, and at the end back to zero. That drawing is the envelope. Attack: how long the climb from silence to the peak takes. Decay: how long the fall from the peak to the sustain level takes. Sustain: a level, not a time, where the sound sits while you hold the key. Release: how long the fade takes after you lift your finger. The brain recognizes an instrument from the first milliseconds: the attack matters most.”

SLIDE 29 • WHERE • AMP ENVELOPE

DEMO: the same note with zero attack, then a long attack; then short vs long release.

0:36–0:37 04-03 TWO ENVELOPES ON FOURM PA

SLIDE 30 • TWO ENVELOPES

SAY: “The Fourm has two. The amp envelope drives the amplifier: the note’s volume contour. The filter envelope drives the cutoff, by however much ENV AMT in the filter section allows: the brightness contour. Same shape, two jobs: how loud, how bright. Watch out for ENV AMT: it’s bipolar, zero at twelve o’clock, plus to the right, minus to the left. Left at twelve, the filter envelope does nothing, the classic trap.”

SLIDE 31 • WHERE • FILTER ENVELOPE › ENV AMT • DECAY

DEMO: “the pluck knob” (30 s): saw, cutoff at half, env amt at three o’clock, filter decay from long to short: “the shorter the decay, the harder the sound plucks.”

0:37–0:38 04-04 BASS vs PAD

SLIDE 32 • BASS VS PAD

SAY: “The same oscillators, two instruments. Bass: instant attack, short decay, low sustain, short release, a pluck that bites. Pad: slow attack, no decay, sustain at maximum, long release, a sound that blooms. And three kitchen rules: with sustain at maximum, the decay has nowhere left to fall and does nothing; with sustain at zero, the note dies while you hold it, that’s the pluck; a filter envelope slower than the amp envelope gives you the pad that opens up. The amp is the gate: whatever the filter does while the amp is still closed, you never hear, a fast filter sweep under a slow amp attack, or a long filter release after a short amp release.”

0:38–0:43 04-05 TRY IT You walk the tables, one ear on every station

SAY: “Headphones on. Partner A builds the bass, B builds the pad, then show each other. Bass, from the module 2 sound (NOISE off, KEY TRACK on, RESONANCE nine o’clock): amp envelope: attack zero, decay ten o’clock, sustain eight o’clock, release eight o’clock. Filter envelope: attack zero, decay ten o’clock, sustain minimum, release eight o’clock. In the filter: ENV AMT to three o’clock, from twelve toward the right!, and cutoff at nine o’clock. Short notes, low. Then shorten the filter decay: more pluck. Pad, partner B: amp envelope, attack twelve o’clock, decay zero, sustain maximum, release two o’clock. Filter envelope, attack two o’clock, decay two, sustain three, release two: slower than the amp, on purpose. Cutoff at twelve o’clock, ENV AMT at two o’clock, again from twelve toward the right. Osc B: SAW on, octave zero, OSC B up in the mixer, FREQUENCY nudged a little. Hold a chord and let it bloom. The positions are starting points, not rules; listen.”

THEY DO: (A) BASS: AMP ENV **attack** 0 • **decay** 10 o’clock • **sustain** 8 o’clock • **release** 8 o’clock; FILTER ENV **attack** 0 • **decay** 10 o’clock • **sustain** min • **release** 8 o’clock; FILTER **env amt** 3 o’clock • **cutoff** 9 o’clock; short notes; then FILTER ENV **decay** shorter. Swap. (B) PAD: AMP ENV **attack** 12 o’clock • **decay** 0 • **sustain** max • **release** 2 o’clock; FILTER ENV **attack** 2 • **decay** 2 • **sustain** 3 • **release** 2 o’clock; FILTER **cutoff** 12 o’clock • **env amt** 2 o’clock; OSC B **saw** on, octave 0, **osc b** up, **frequency** slightly off-centre; hold a chord. Note the positions you liked, or photograph the panel.

0:43–0:44 04-06 KEY TERMS

SLIDE 33 • KEY TERMS

SAY: “Attack, decay, sustain, release, env amount, velocity. The envelope is what makes a synth sound like an instrument instead of a signal. And the beautiful part: four knobs describe almost every instrument on Earth.”

0:44–0:53

MODULE 05 · LFO & MODULATION

0:44–0:45

05·01 COVER Headphones down

SLIDE 34 · COVER

SAY: “Envelopes give the note its shape; modulation keeps it alive. A sound that stands still sounds like a signal; every sound in nature moves. Modulation is the verb of synthesis: something moves something else.”

0:45–0:47

05·02 THE LFO PA

SLIDE 35 · THE LFO

SAY: “LFO: low frequency oscillator. An oscillator too slow to hear as a note: you hear what it moves. Sent to pitch: vibrato, the violinist rocking a finger on the string. Sent to cutoff: wah. Sent to the amplifier: tremolo, the volume rocked twice a second. The FREQUENCY knob is the speed, SYNC ties it to the tempo, and the shapes can be combined: triangle for smooth vibrato, square for trills, sample and hold for random, stepped motion. Musical vibrato is small and about six or seven times a second; speed it up and it's not vibrato anymore, it's a growl.”

SLIDE 36 · WHERE · LFO

DEMO: (45 s): one LFO → cutoff, FREQUENCY from very slow (ambient sweep) to 1–2 Hz (wah) to 15 Hz (growl); then the same LFO → pitch (vibrato) → amp (tremolo).

0:47–0:49

05·03 THE MODULATION SECTION

SLIDE 37 · THE MODULATION SECTION

SAY: “Look at the left side of the panel. Three sources: filter env, osc B, and the LFO, each with a ROUTE button and an amount knob. Eight destinations on the right: freq A, freq B, PW A, PW B, cutoff, amp, LFO freq, LFO amount. Press ROUTE once: red LED, the direct bus: the modulation is always on. Press it twice: blue, the wheel bus: the modulation only appears as you raise the mod wheel. The wheel becomes the modulation's volume; without it, a vibrato would be permanently on, which isn't musical. Destinations work the same: one press, red; two, blue; three, both, purple. And the trap of the evening, once more: the amount knobs are bipolar, zero at twelve o'clock. Nothing happening? Turn from twelve toward the right.”

SLIDE 38 · WHERE · MODULATION SECTION

DEMO: vibrato on the wheel, ROUTE and destinations on blue; raise the wheel slowly.

0:49–0:52

05·04 TRY IT You walk the tables, checking LED colors

SAY: “Headphones on, on the pad from module four. In the LFO: TRI lit, the other shapes dark, FREQUENCY at ten o'clock. In Modulation, look at the LEDs: the ROUTE under LFO, FREQ A and FREQ B are already blue, the Basic Program wired the vibrato to the wheel for you; if one is dark or red, press it until it's blue. Push the mod wheel all the way up, hold a chord and turn the LFO's AMOUNT from twelve toward the right until you hear the vibrato. Wheel down, wheel up: vibrato under your control. Keep the wheel up and swap. Partner: press FREQ A and FREQ B until they go dark, press PW A until it's blue, the pulse breathes. Then CUTOFF until it's blue, wah. Faster, more: go too far, then bring it back.”

THEY DO: (A) LFO **tri** lit (others off), **frequency** 10 o'clock → MOD: **route** (lfo) blue, **freq a** blue, **freq b** blue (press until blue if not) → mod wheel up → **lfo** amount from 12 o'clock to the right → wheel down / up. Swap, wheel up. (B) **freq a/freq b** until dark → **pw a** until blue → plays; then **cutoff** until blue → **frequency** and amount higher, then back.

0:52–0:53

05·05 KEY TERMS

SLIDE 39 · KEY TERMS

SAY: “LFO, rate, vibrato, tremolo, PWM, modulation bus, mod wheel. And a promise kept: the PWM from module two: you just made it with your own hands.”

0:53–1:02

MODULE 06 · VOICES & UNISON

0:53–0:54

06·01 COVER Headphones down

SLIDE 40 · COVER

SAY: “So far we’ve built one voice. The Fourm has four, each with its own oscillators, filter and envelopes. This module is about how we spend them: spread across a chord, stacked onto one huge note, or replayed as a memorized chord from a single key. And about the loop that fattens everything.”

0:54–0:55

06·02 POLYPHONY PA

SLIDE 41 · POLYPHONY

SAY: “A mono synth plays one note at a time. The Fourm plays four; the fifth steals the oldest. A ten-second story: the Oberheim Four Voice, 1975, was four complete voice modules, each with its own oscillators, filter and envelopes, tuned by hand, under a single keyboard. In unison, one of the most overwhelming sounds ever made. The Fourm is the same idea, in a four-kilogram box. HOLD sustains up to four notes even after your hands leave.”

DEMO: five notes, you hear the steal; then HOLD.

0:55–0:57

06·03 UNISON PA

SLIDE 42 · UNISON

SAY: “Unison: every chosen voice, from one to four, hold UNISON and turn SELECT, on a single note. The Fourm goes monophonic and, with four voices, huge. On its own it’s just louder; the magic is the detune: parameter ten, Unison Detune, a few cents between the voices and the tone turns wide, alive, pulsing. When it sounds huge, pull it back ten percent: that’s usually where the good sound lives.”

SLIDE 43 · WHERE · UNISON + SELECT

DEMO: unison with no detune, then detune raised slowly until it shimmers, then too far, then back.

0:57–0:58

06·04 CHORD MEMORY PA

SLIDE 44 · CHORD MEMORY

SAY: “Hold a chord, memorize it, and from then on every key plays the whole chord, transposed. One finger, full harmony; for chord stabs it’s gold. It saves with the program. This is how the rave stabs of the early nineties were played: one finger on a synth with chord memory, and the whole hall thought it was a keyboard player.”

SLIDE 45 · WHERE · CHORD MEMORY

DEMO: at the presenter’s synth, with the real steps: (UNISON → SELECT to the middle row → Chord → UNISON → hold the chord → UNISON): memorize a minor chord and walk it chromatically with one finger. Participants just listen; the steps are in the student handbook, under Level 2.

0:58–0:59

06·05 FEEDBACK PA

SLIDE 46 · FEEDBACK

SAY: “The promise from module one: the loop. A mixer channel feeds the Fourm’s output back into the mix, with soft-clipping diodes along the way. Press FEEDBACK (it comes in at one hundred), hold and turn SELECT for the amount. A little means warmth, a lot means snarl. Unison plus feedback is the biggest bass the Fourm can make. This started as a hack: on the original Minimoog people ran a cable from the spare main output back into the external input and turned that input up until the mixer overloaded. Half a century later, the Fourm has that loop built in, with the diodes that keep it thick instead of screaming.”

SLIDE 47 • WHERE • MIXER › FEEDBACK + SELECT

DEMO: on a held bass, feedback from clean to aggressive and back. **Optional (30 s):** “Glass Echo”, F117: self-oscillating filter + feedback: a tuned echo appears on the release.

0:59–1:01 06-06 TRY IT You walk the tables

SAY: “Headphones on. Play five notes at once and count what you hear. Hold UNISON, SELECT to four voices, play a line: one huge note. Press PARAM, go to ten (Unison Detune), press SELECT, turn to about forty, PROGRAM to exit: the tone widens. Press FEEDBACK, hold and turn SELECT down to sixty: snarl without chaos. Swap: your partner turns unison off, plays a chord, presses HOLD and lifts both hands.”

THEY DO: (A) 5 notes → hold **unison** + **select** = 4 → line → **param** → 10 → press **select** → ~40 → **program** → **feedback** on → hold **feedback** + **select** → ~60. Swap. (B) **unison** off → chord → **hold** → hands up.

1:01–1:02 06-07 KEY TERMS

SLIDE 48 • TERMS

SAY: “Polyphony, voice, unison, detune in cents, chord memory, hold, feedback, soft clipping. The practical advice, once more: after it sounds big, pull everything back ten percent.”

1:02–1:10

MODULE 07 • EXPRESSION

1:02–1:03 07-01 COVER Headphones down

SLIDE 49 • COVER

SAY: “Until now we’ve been building the sound; from here on we play it. Envelopes and LFOs make every note the same; what turns that into music is what your hands add. Velocity reads how fast you strike, aftertouch reads how hard you keep pressing, glide ties the notes together. The Fourm’s Tactive keyboard feels the pressure under each individual key.”

1:03–1:04 07-02 VELOCITY PA

SLIDE 50 • VELOCITY

SAY: “On real instruments, louder notes also carry more harmonics: that’s why velocity goes to the filter, not just the volume. Each envelope has a VELOCITY button: press it and how hard you play scales that envelope’s depth. Velocity on the filter envelope: soft stays dark, hard bites. Five response curves in Global, from Soft+ to Hard+, match the keyboard to your touch.”

SLIDE 51 • WHERE • VELOCITY

DEMO: the same phrase pianissimo, then forte, with velocity on the filter.

1:04–1:05 07-03 POLY AFTERTOUCH PA

SLIDE 52 • POLY AT

SAY: “The Fourm’s signature. After the note is down, extra pressure becomes a continuous modulator, on every key. Press into a single note of a chord and only that one blooms, like blowing harder into a flute: louder, brighter. On most keyboards aftertouch is mono, it moves all the voices at once; per-note was a rarity, the Yamaha CS-80, the Prophet T8, and the Fourm has it natively.”

DEMO: held chord, press into each note in turn.

1:05–1:06 07-04 PRESSURE ROUTING

SLIDE 53 • AT ROUTING

SAY: “Where you send the pressure: six destinations (freq A, freq B, filt, amp, LFO freq, LFO amount), chosen with the three DEST buttons, two per button, and a single AMOUNT knob, plus or minus. Cutoff is the classic: press for brightness, let go to relax; with negative amount, it closes. Pitch is for the brave.”

SLIDE 54 • WHERE • AFTERTOUCH

DEMO: switch the destination live between filt and freq.

1:06–1:07 07-05 GLIDE PA

SLIDE 55 • STEPS VS GLIDE

SAY: “Glide, or portamento: the pitch slides from the previous note to the next instead of jumping. Press GLIDE and set the RATE: on with rate at zero it does nothing. Subtle almost always sounds more musical than spectacular: it’s the difference between a keyboard melody and a phrase that sounds like a voice.”

DEMO: the same bass phrase without glide, then with. “The modes, four of them, are in the student handbook.”

1:07–1:09 07-06 TRY IT You walk the tables

SAY: “Headphones on. In FILTER ENVELOPE press VELOCITY; play soft, then hard. In AFTERTOUCH: press the DEST button under FILT/AMP until only FILT stays lit, AMOUNT at two o’clock; hold a three-note chord and press into the top note only. Now AMOUNT at ten o’clock: pressure closes the filter. Swap. Partner: GLIDE on, RATE at ten o’clock, a bass line: the notes slide. Then move the pressure to FREQ A and FREQ B, with a small amount: a bend, under one finger.”

THEY DO: (A) FILTER ENV **velocity** on → soft/hard → AFTERTOUCH **dest** (FILT/AMP) until only **filt** → **amount** 2 o’clock → chord, pressure on the top note → **amount** 10 o’clock. Swap. (B) **glide** on → **rate** 10 o’clock → bass line → **dest** (FREQ A/FREQ B) → small amount → bend under a finger.

1:09–1:10 07-07 KEY TERMS

SLIDE 56 • TERMS

SAY: “Velocity, poly aftertouch, pressure routing, Tactive, glide, portamento, legato. Invite your partner to feel the press inside a chord: the sensation sticks better than the definition.”

1:10–1:17

MODULE 08 • ARPEGGIATOR & SEQUENCER

1:10–1:11 08-01 COVER Headphones down

SLIDE 57 • COVER

SAY: “Patterns out of held notes: harmony becomes rhythm. And behind the arpeggiator, the Fourm hides a 64-step sequencer. A mouse is not a musical instrument: that’s why the arpeggiator lives on the panel.”

1:11–1:12 08-02 HOLD A CHORD PA

SLIDE 58 • HOW IT WORKS

SAY: “Press ON, hold the chord, the synth plays the pattern: up, down, up-down, random, or in the order you pressed, across one to three octaves; hold MODE or OCTAVE and turn SELECT. HOLD keeps it running hands-free. And a trick: change how many notes you’re holding and the rhythm changes: it sounds like a mistake until you embrace it.”

SLIDE 59 • WHERE • ARPEGGIATOR • CLOCK

DEMO: arp on a triad, switch directions live; then four notes, then two.

1:12–1:13 08-03 CLOCK & SYNC PA

SLIDE 60 • CLOCK

SAY: “One clock drives everything: tap TAP TEMPO four times, or hold it and turn SELECT for the BPM. CLK DIV picks the grid, quarters, eighths, sixteenths, triplets, and the LFO’s SYNC follows the same clock. If the arp won’t start: Clock Mode in Global is on In, waiting for external clock.”

DEMO: the same arpeggio through three divisions, without stopping the pattern.

1:13-1:14 08-04 NOTE & MOD SEQ

SLIDE 61 • SEQUENCER

SAY: “Beyond arpeggios: up to 64 recorded steps: notes, chords, rests, ties, even a glide per step, triggered and transposed from the keyboard; or a sequence that moves a parameter instead of playing notes: cutoff, pulse width, feedback. A sequencer in disguise, and a story for another workshop.” **The image:** “a sequencer is, at its origin, a row of eight knobs walked through by a clock: how far you turn each one draws the melody; that’s how the bass of ‘I Feel Love’ was made, on a Moog modular.”

1:14-1:16 08-05 TRY IT You walk the tables · leave your own arp running slowly in the background

SAY: “Headphones on. Arpeggiator ON, hold C–E–G, press HOLD (in the UNISON section) and lift your hands. Tap TAP TEMPO four times at your own tempo. Hold CLK DIV and turn SELECT to sixteenths, then back to eighths. Ride the CUTOFF as far as it goes. Swap. Partner: hold MODE and turn SELECT to Random; hold OCTAVE and turn SELECT to 2. Then a ‘wrong’ chord, and let it repeat: give it eight bars.”

THEY DO: (A) ARP on → C–E–G → hold → tap tempo/bpm ×4 → hold clk div + select → 1/16 → 1/8 [check at home: hold + select vs. repeated presses; it’s also Param 19] → cutoff. Swap. (B) hold mode + select → Random → hold octave + select → 2 → a “wrong” chord, 8 bars.

1:16-1:17 08-06 KEY TERMS

SLIDE 62 • TERMS

SAY: “Arpeggiator, hold, clock, BPM, division, note and mod sequencer. Leave the arps running quietly while we recap: the room stays in rhythm.”

1:17-1:30 MODULE 09 · WRAP-UP · Q&A · FREE PLAY

1:17-1:18 09-01 THE ROAD, AGAIN Headphones down

SLIDE 63 • THE ROAD AGAIN

SAY: “This is the road: oscillator, mixer, filter, envelopes, LFO and modulation, voices, expression, rhythm. Every synth you’ll meet from now on is the same road in different clothes. If you know the road, you know where to look.”

1:18-1:20 09-02 CHALLENGE You walk the tables

SAY: “Last exercise, two minutes. From Basic Program, rebuild the bass from module four from memory: A builds, B watches the panel and prompts, then swap. When you’re done, write your settings down or photograph the panel: a sound on paper survives, one in memory rarely does.”

THEY DO: (A) hold transpose down + program → the BASS recipe from module 04, from memory. Swap. (B) checks, then rebuilds it too.

1:20-1:25 09-03 Q&A

SAY: “Questions. Anything: from ‘what’s that knob’ to ‘how do I make the sound from track X’” **Your moderating rule:** with every answer, tie the term back to the demo where they heard it. When someone describes a sound, ask for concrete words: bright/dark, thin/fat, pluck/pad, static/moving.

1:25-1:26 09-04 THANK YOU

SLIDE 64 • THANK YOU

SAY: “Thank you. Zeedo, thank you for making this possible; Sequential, thank you for the instruments. The student handbook is yours, in English or Romanian: the same diagrams, all the theory. You’ll find us at Sintezaur, [where], and that’s where we announce the next workshop. Until then: the synths are yours for another ten minutes.”

1:26-1:30 09-05 FREE PLAY

SAY: “Ten free minutes. One suggestion, if you want one: a single sound, sixty seconds: starts quiet, grows, hits a peak, comes down. The cutoff and the resonance are the story.” You walk the tables, going to raised hands; group photo at the end.

The Glossary annex is never presented on screen; it lives only in the student handbook.

AFTER

- **Repack:** every Fourm in its own box, with the original packaging, power supply alongside; headphones and splitters counted (20 + 10 from the kit, Zeedo's separately); stands, power strips, extension cords; the return checklist signed.
- **Take notes within 10 minutes, while it's fresh:** what worked, what ran long, what questions came up, for v2 of the course.

PRESENTER CHEAT SHEET

The exercises + the challenge, on one page

Control names are the ones on the panel (**bold**). “A → swap → B” = partner A is at the controls first, then they swap.

MODULE	MINUTE	THEY DO (condensed)	WHO DRIVES
01 • SIGNAL FLOW	0:10–0:12	A: osc a to zero → silence → back; Osc B saw on, osc b up (saw lit). B: repeats; both trace the path across the panel with a finger.	A, then B
02 • OSCILLATORS	0:19–0:22	A: Osc A pulse on (saw + pulse) → hold pulse pw + select toward the edge and back → Osc B tri on, octave -1, osc b up → B's frequency slightly off-centre → back. B: noise on → hold noise + select ≈ 30 → plays; then A's steps. The sound stays on the synth (raw material for the bass).	A → swap → B
03 • FILTER	0:29–0:32	A: cutoff down → up; resonance at 3 o'clock, sweep. B: main slightly down → resonance max → cutoff a bit closed → single notes; hold key track + select → Full; then resonance 9 o'clock, cutoff 12 o'clock.	A → swap → B
04 • ENVELOPES	0:38–0:43	A: BASS: AMP ENV attack 0 · decay 10 · sustain 8 · release 8 o'clock; FILTER ENV attack 0 · decay 10 · sustain min · release 8; FILTER env amt 3 o'clock · cutoff 9 o'clock; short notes; then FILTER ENV decay shorter. B: PAD: AMP ENV attack 12 · decay 0 · sustain max · release 2 o'clock; FILTER ENV attack 2 · decay 2 · sustain 3 · release 2 o'clock; cutoff 12 · env amt 2 o'clock; Osc B saw , octave 0, osc b up, frequency slightly off-centre; held chord. Good settings get written down, or the panel gets photographed.	A bass → swap → B pad
05 • LFO & MODULATION	0:49–0:52	A: LFO tri lit (others off), frequency 10 o'clock → MODULATION route (LFO), freq a , freq b all blue (press until blue if not) → mod wheel up → LFO amount from 12 o'clock to the right → wheel down / up. B, wheel up: freq a / freq b until dark → pw a until blue → plays; then cutoff until blue → frequency and amount higher, then back.	A → swap → B
06 • VOICES & UNISON	0:59–1:01	A: 5 notes → hold unison + select = 4 → line → param → 10 → press select → ≈ 40 → program → feedback on → hold feedback + select → ≈ 60. B: unison off → chord → hold → hands up.	A → swap → B
07 • EXPRESSION	1:07–1:09	A: FILTER ENV velocity on → soft / hard → AFTERTOUCH dest (FILT / AMP) until only FILT → amount 2 o'clock → chord, pressure on the top note → amount 10 o'clock. B: glide on → rate 10 o'clock → bass line → dest (FREQ A / FREQ B) → small amount → bend under a finger.	A → swap → B
08 • ARPEGGIATOR & SEQ	1:14–1:16	A: ARP on → C–E–G → hold → tap tempo bpm × 4 → hold clk div + select → 1/16 → 1/8 → cutoff . B: hold mode + select → Random → hold octave + select → 2 → a “wrong” chord, 8 bars.	A → swap → B
09 • CHALLENGE	1:18–1:20	From Basic Program (hold transpose down + program), rebuild the bass from 04 from memory: A builds, B watches the panel and prompts; then the other way. Settings get written down.	A → swap → B

THE TRAPS (BACK OF THE PAGE)

- **If you hear nothing:** headphones in the splitter? **main** at 12 o'clock? at least one mixer source up? **cutoff** closed? **hold** lit from the last exercise? mod wheel down on a blue route? Then Basic Program (hold **transpose down** + **program**), it resets the mixer, the cutoff and the amounts. Still silent: Global, Local Control = On. Knobs that do nothing: Global 4, Pot Mode is not Jump.
- **Self-oscillation comes in sudden and loud:** **main** slightly down before **resonance** hits maximum. Announce it every time.
- **The amounts are bipolar** (MODULATION, **env amt** and the AFTERTOUC amount): 12 o'clock = nothing. Say "from twelve toward the right", never just "raise the amount".
- **sustain at maximum = the decay does nothing.** When someone "can't hear the decay", look at the sustain.
- **A short press of program is harmless** (it closes PARAM); turning **select** or pressing INC/DEC on the program screen loads another program and loses the sound. Remind them at every exercise.
- **Chord memory is demo-only**, at your synth (steps through the Unison menu), not at the tables, not during the exercise.
- **Aftertouch:** say what is certainly true, „press harder → brighter (FILT) / bend (FREQ), on the pressed note“, and leave what the LFO shares between voices to the Fourm manual.
- **CLK DIV:** hold + **select** or repeated presses [check at home]; it's also Param 19.
- **Headphones:** on while they play, one ear free while you talk; swap at every exercise, so the panel passes through both pairs of hands.